

WT LAB 2

Study of Various Types of Sensors and Display Devices

Aim :-

To study and demonstrate the interfacing of various sensors (e.g., DHT-11/22, HC-SR04, MFRC522, PIR Sensor) and display devices using Arduino/Raspberry Pi. Additionally, to select a mini-project topic based on the concepts learned.

Theory :-

- Sensors are critical components in wireless and IoT systems, enabling the collection of real-world data for processing and analysis. This experiment focuses on understanding the working principles of commonly used sensors and their integration with microcontrollers like Arduino or Raspberry Pi. Display devices are used to visualize the data collected by the sensors.

List of Sensors and Display Devices

1. **DHT-11/22:** Temperature and humidity sensor.
2. **HC-SR04:** Ultrasonic sensor for distance measurement.
3. **MFRC522:** RFID module for contactless communication.
4. **PIR Sensor:** Passive Infrared sensor for motion detection.
5. **Display Devices:** LCD, OLED, or 7-segment displays for output visualization.

Experiment Setup

Components Required

- Arduino Uno/Raspberry Pi
- DHT-11/22 Sensor
- HC-SR04 Ultrasonic Sensor
- MFRC522 RFID Module

- PIR Sensor
- LCD/OLED Display
- Jumper Wires
- Breadboard
- Power Supply

Interfacing and Demonstration

1. DHT-11/22 Sensor (Temperature and Humidity)

Working Principle: Measures temperature and humidity using a capacitive humidity sensor and a thermistor.

Interfacing with Arduino:

- Connect VCC to 5V, GND to GND, and Data Pin to a digital pin (e.g., D2).
- Use the DHT library to read data.

Code Example:

```
#include <DHT.h>
#define DHTPIN 2
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(9600);
  dht.begin();
}

void loop() {
  float h = dht.readHumidity();
  float t = dht.readTemperature();
  Serial.print("Humidity: "); Serial.print(h);
  Serial.print(" %, Temperature: "); Serial.print(t); Serial.println(" °C");
  delay(2000);
}
```

2. HC-SR04 Ultrasonic Sensor (Distance Measurement)

Working Principle: Uses ultrasonic waves to measure distance by calculating the time taken for the wave to reflect back.

Interfacing with Arduino:

- Connect VCC to 5V, GND to GND, Trig to D9, and Echo to D10.

Code Example:

```
#define trigPin 9
#define echoPin 10

void setup() {
  Serial.begin(9600);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
}

void loop() {
  long duration, distance;
  digitalWrite(trigPin, LOW); delayMicroseconds(2);
  digitalWrite(trigPin, HIGH); delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
  distance = (duration / 2) / 29.1;
  Serial.print("Distance: "); Serial.print(distance); Serial.println(" cm");
  delay(500);
}
```

3. MFRC522 RFID Module

Working Principle: Uses radio frequency to read and write data on RFID tags.

Interfacing with Arduino:

- Connect VCC to 3.3V, GND to GND, RST to D9, SDA to D10, MOSI to D11, MISO to D12, and SCK to D13.

Code Example:

```
#include <SPI.h> #
include <MFRC522.h>
#define RST_PIN 9
#define SS_PIN 10
MFRC522 mfrc522(SS_PIN, RST_PIN);
```

```
void setup() {
  Serial.begin(9600);
  SPI.begin();
  mfrc522.PCD_Init();
  Serial.println("Scan RFID Tag...");
}

void loop() {
  if (mfrc522.PICC_IsNewCardPresent() && mfrc522.PICC_ReadCardSerial()) {
    Serial.print("UID: ");
    for (byte i = 0; i < mfrc522.uid.size; i++) {
      Serial.print(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " ");
      Serial.print(mfrc522.uid.uidByte[i], HEX);
    }
    Serial.println();
    mfrc522.PICC_HaltA();
  }
}
```

4. PIR Sensor (Motion Detection)

Working Principle: Detects infrared radiation emitted by objects in its field of view.

Interfacing with Arduino:

- Connect VCC to 5V, GND to GND, and OUT to D2.

Code Example:

```
#define pirPin 2
void setup() {
  Serial.begin(9600);
  pinMode(pirPin, INPUT);
}

void loop() {
  int motion = digitalRead(pirPin);
  if (motion == HIGH) {
    Serial.println("Motion Detected!");
  } else {
    Serial.println("No Motion");
  }
  delay(1000);
}
```

5. Display Devices (LCD/OLED)

Interfacing with Arduino:

- Connect VCC to 5V, GND to GND, SDA to A4, and SCL to A5 for I2C communication.

Code Example:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h> LiquidCrystal_I2C
lcd(0x27, 16, 2);

void setup() { lcd.begin();
  lcd.backlight();
  lcd.print("Hello, World!");
}

void loop() { lcd.setCursor(0,
  1);
  lcd.print("Temp: 25 C"); delay(1000);
}
```

Conclusion :-

The various types of sensors, types of displays, their functioning, how they interface with Arduino, and example code to use them, was studied, and understood.

Sensor Type	Principle of Operation	Measured Parameter	Accuracy	Response Time	Application Areas	Cost
Temperature Sensor	Change in resistance, voltage, or current due to temperature	Temperature	High ($\pm 0.1^\circ\text{C}$ for RTD)	Fast (ms to seconds)	HVAC, Industrial, Medical	Low to Medium
Pressure Sensor	Deformation of diaphragm, piezoelectric effect	Pressure	High ($\pm 0.1\%$ FS)	Fast (ms)	Automotive, Aerospace, Industrial	Medium
Proximity Sensor	Inductive, capacitive, or optical detection of nearby objects	Presence/Absence of object	Medium	Very fast (μs to ms)	Robotics, Manufacturing, Automation	Medium
Infrared (IR) Sensor	Emission and detection of infrared radiation	Heat, Motion	Medium	Fast (ms)	Security, Remote Controls, Thermal Imaging	Low
Ultrasonic Sensor	Emitting and receiving high-frequency sound waves	Distance, Obstacle Detection	Medium	Fast (ms)	Robotics, Automotive, Level Measurement	Medium
LIDAR Sensor	Time-of-flight measurement using laser beams	Distance, 3D Mapping	Very High	Fast (μs to ms)	Autonomous Vehicles, Surveying, Mapping	High
Hall Effect Sensor	Change in voltage due to magnetic field	Magnetic Field, Position	High	Very fast (μs)	Automotive, Industrial, Consumer Electronics	Low
Gas Sensor	Chemical reaction causing change in resistance	Gas Concentration	Medium	Moderate (seconds to minutes)	Environmental Monitoring, Safety Systems	Medium
Accelerometer	MEMS-based measurement of acceleration	Acceleration, Vibration	High	Fast (ms)	Smartphones, Automotive, Industrial	Medium
Gyroscope	MEMS or optical-based angular velocity detection	Angular Velocity	High	Fast (ms)	Aerospace, Consumer Electronics, Automotive	Medium
Humidity Sensor	Capacitive or resistive change due to moisture	Humidity	Medium to High	Moderate (seconds)	Weather Stations, Industrial, Agriculture	Low